



Gowin EDP Encoder Decoder IP User Guide

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Revision History

Date	Version	Description
12/13/2023	1.0E	Initial version published.
02/23/2024	1.1E	Parameters updated.
03/29/2024	2.0E	● IP name updated to EDP Encoder Decoder. ● Descriptions of EDP Encoder port parameter calculation updated. ● Interface screenshots updated.
06/07/2024	2.1E	● EDP Decoder ports updated. ● Descriptions for sideband control added. ● Timing diagrams updated.
01/17/2025	2.2E	● EDP Encoder and Decoder ports added. ● Interface screenshots updated.
07/18/2025	2.3E	● IP features updated. ● New ports added to EDP Encoder and EDP Decoder. ● Parameter calculation descriptions for EDP Encoder ports updated. ● Configuration parameters for both EDP Encoder and Decoder updated. ● Interface screenshots updated.
09/12/2025	2.4E	EDP Decoder ports updated.
01/30/2026	2.5E	● Updated eDP Decoder port descriptions. ● Added descriptions for the RX HDCP Enable and RX Secondary Data Packet Enable parameters.
06/26/2026	2.6E	● Updated EDP Encoder and EDP Decoder ports. ● Updated EDP Encoder and EDP Decoder parameters.

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1 About This Guide

1.1 Purpose

The purpose of Gowin EDP Encoder Decoder IP is to help you learn the features and usage of Gowin EDP Encoder Decoder IP by providing the descriptions of features, functions, ports, timing, GUI and reference design, etc. The software screenshots and the supported products listed in this manual are based on Gowin Software V1.9.12.03 (64-bit). As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI website. You can find the related documents at www.gowinsemi.com:

- [DS981, GW5AT series of FPGA Products Data Sheet](#)
- [DS1103, GW5A series of FPGA Products Data Sheet](#)
- [DS1239, GW5AST series of FPGA Products Data Sheet](#)
- [DS1105, GW5AS series of FPGA Products Data Sheet](#)
- [DS1108, GW5AR series of FPGA Products Data Sheet](#)
- [DS1118, GW5ART series of FPGA Products Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
AUX CH	Auxiliary Channel
DE	Data Enable
DP	DisplayPort
eDP	Embedded DisplayPort

Terminology and Abbreviations	Meaning
FPGA	Field Programmable Gate Array
HPD	Hot Plug Detect
HS	Horizontal Synchronization
IP	Intellectual Property
RGB	R(Red) G(Green) B(Blue)
SerDes	Serializer/Deserializer
TU	Transfer Unit
VESA	Video Electronics Standards Association
VS	Vertical Synchronization

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

2.1 Overview

DisplayPort (DP) is a digital video interface standard developed by the PC and chip manufacturing industry in collaboration with the Video Electronics Standards Association (VESA). This interface is free from certification and licensing fees. It is primarily used for connecting video sources to display devices and supports the transmission of audio, USB, and other forms of data.

The Embedded DisplayPort (eDP) protocol is an extension of the DP protocol specifically designed for embedded systems. eDP is fully compatible with the DP protocol, and it is commonly used in applications where DP is applied in embedded architectures. It is frequently utilized as the interface for liquid crystal displays (LCDs) in tablets and laptop computers.

The EDP Encoder IP is designed to receive parallel video signals, encode them according to the eDP protocol, and output the encoded data to SerDes for serialization. The EDP Decoder IP is designed to receive data that has been deserialized by SerDes, and then decode it based on the eDP protocol to generate parallel video signals.

Table 2-1 Gowin EDP Encoder Decoder IP

Gowin EDP Encoder Decoder IP	
Logic Resource	Please refer to Table 2-2 and Table 2-3.
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.9(64-bit) and above)

Note!

For the devices supported, you can click [here](#) to get the information.

2.2 Features

- Supports 1, 2, or 4 lanes
- Supports 8B10B encoding and decoding
- Supports Word Alignment
- Supports 1.62/2.7/5.4/8.1 Gbps/lane
- Supports 16/32/64 bit widths

2.3 Resource Utilization

EDP Encode and Decode can be implemented by Verilog. Its performance and resource utilization may vary when the design is employed in different devices, or at different densities, speeds, or grades. Take GW5AST series of FPGA as an instance, and the resource utilization is as shown in Table 2-2 and Table 2-3.

Table 2-2 EDP Encoder Resource Utilization

Device	Speed Grade	Name	Resource Utilization	Note
GW5AST-138	C1/I0	LUT	2244	4 lanes, 5.4Gbps/lane
		REG	3721	
		BSRAM	8	

Table 2-3 EDP Decoder Resource Utilization

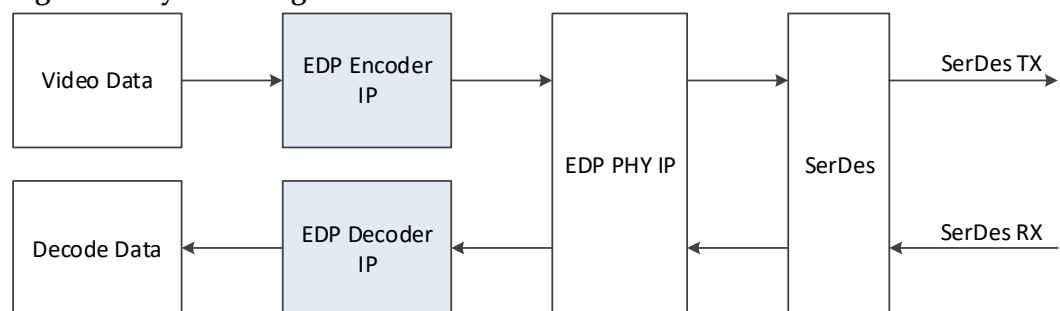
Device	Speed Grade	Name	Resource Utilization	Note
GW5AST-138	C1/I0	LUT	7008	4 lanes, 5.4Gbps/lane
		REG	7631	
		BSRAM	0	

3 Functional Description

3.1 System Block Diagram

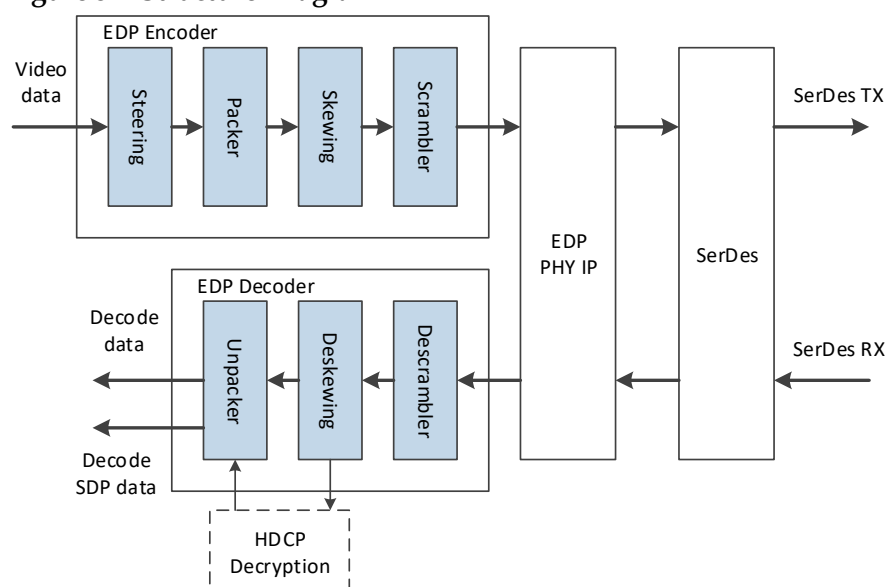
Gowin EDP Encoder Decoder IP includes EDP Encoder IP and EDP Decoder IP, and its system diagram is as shown in Figure 3-1.

Figure 3-1 System Diagram



3.2 Structure Diagram

Figure 3-2 Structure Diagram



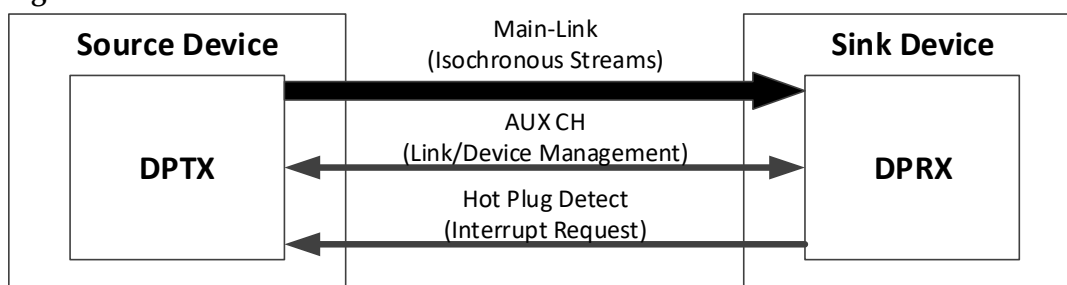
Gowin EDP Encode IP consists of a Steering module, a Packer module, a Skewing module, and a Scrambler module. EDP Decoder IP consists of a Descrambler module, a Deskewing module, and an Unpacker module.

3.3 Working Principle

According to DisplayPort Version 1.4, the complete DP transceiver controller can be broadly divided into two main parts, as shown in

1. Main Link: Includes link data packing/unpacking and SerDes physical layer.
2. AUX CH + HPD: Includes link layer management, auxiliary channel, and hot-plug detect.

Figure 3-3 DP Data Transmission Channel



Gowin EDP Encoder Decoder IP is only involved in the Main Link, specifically in link data packing and unpacking.

Note!

This manual provides a brief introduction to DP protocol-related terms and definitions. For detailed information, please refer to the DisplayPort Version 1.4 protocol.

3.3.1 Main-Link Data Packing/Unpacking

Steering Module

According to the DisplayPort (DP) protocol, the main link can support 1, 2, or 4 lanes. Regardless of the color space and pixel bit depth of the video stream, pixel is steered according to Table 3-1.

Table 3-1 Pixel Steering into Main Link Lanes

# of Lanes	Pixel Steering
One	All pixels to Lane 0
Two	Pixel 2N to Lane 0 Pixel 2N+1 to Lane 1
Four	Pixel 4N to Lane 0 Pixel 4N+1 to Lane 1 Pixel 4N+2 to Lane 2 Pixel 4N+3 to Lane 3

Gowin EDP Encoder IP determines the video input data width based on the number of main link lanes. When the main link has 1 lane, the video

input data width is 24 bits. For 2 lanes, it is 48 bits, and for 4 lanes, it is 96 bits.

Note!

- If the Bits Per Pixel Coefficient is set to 1, the video input data width follows the values mentioned above.
- If the Bits Per Pixel Coefficient is set to 2, the input data width follows the bit width described above multiplied by 2.
- If the Bits Per Pixel Coefficient is set to 4, the input data width follows the bit width described above multiplied by 4.

Packer and Unpacker Modules

According to the DP protocol, the following control symbols are used to framing.

- BS: Blanking Start
- BE: Blanking End, inserted before the first active video in each line.
- FS: Fill Start
- FE: Fill End
- SS: Secondary-data Start
- SE: Secondary-data End
- SR: Scrambler Reset, and the SR symbol is used to reset the LFSR to its initial value.
- BF: Blanking Fill for enhanced framing mode

In the ANSI standard, the correspondence between 8B/10B special K character and the control symbols is as shown in Table 3-2.

Table 3-2 K Character and the Control Symbols

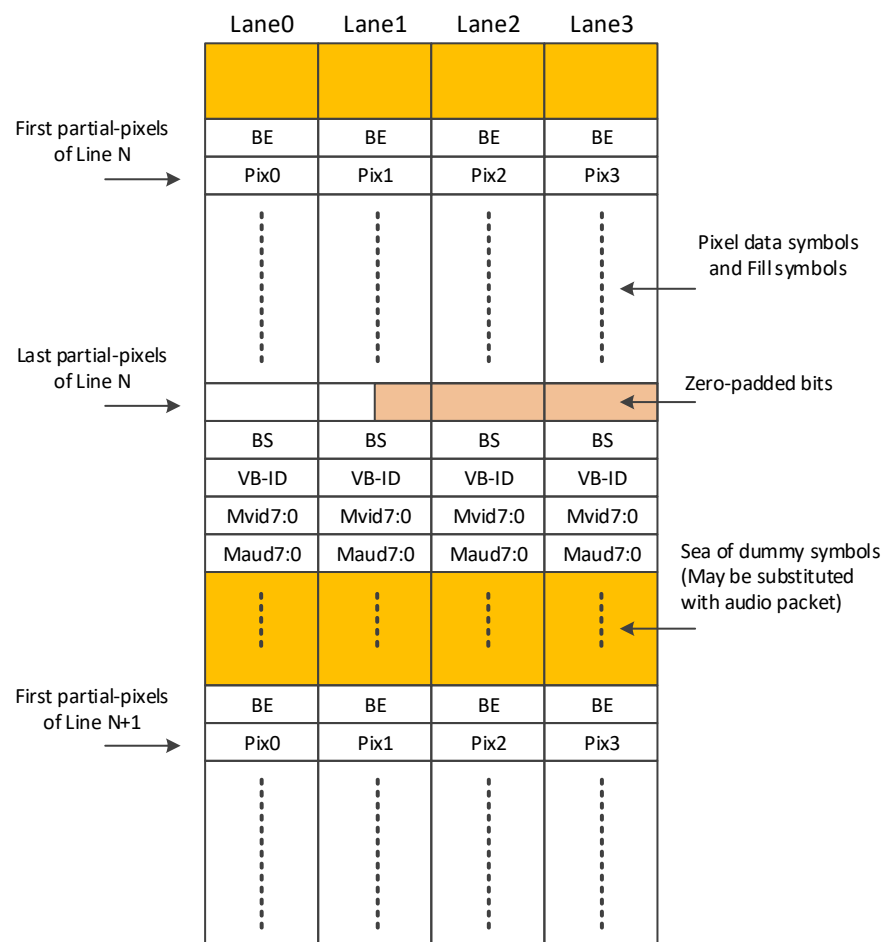
Special Character	Control Symbol in Single Stream Enhanced Mode
K28.0	SR
K28.1	CP
K28.2	SS
K28.3	BF
K28.5	BS
K23.7	FE
K27.7	BE
K29.7	SE
K30.7	FS

According to the DP protocol, the differences between the default framing mode and enhanced framing mode symbols are as shown in Table 3-3.

Table 3-3 Differences between the Default Framing Mode and Enhanced Framing Mode Symbols

Default Framing Mode Symbols	Enhanced Framing Mode Symbols
BS	BS+BF+BF+BS
SR	SR+BF+BF+SR
CPBS(called CP symbol in Enhanced Framing Mode)	BS+CP+CP+BS
CPSR(called BF symbol in Enhanced Framing Mode)	SR+CP+CP+SR
BE	BE(no change)
FS	FS(no change)
FE	FE(no change)
SS	SS(no change)
SE	SE(no change)

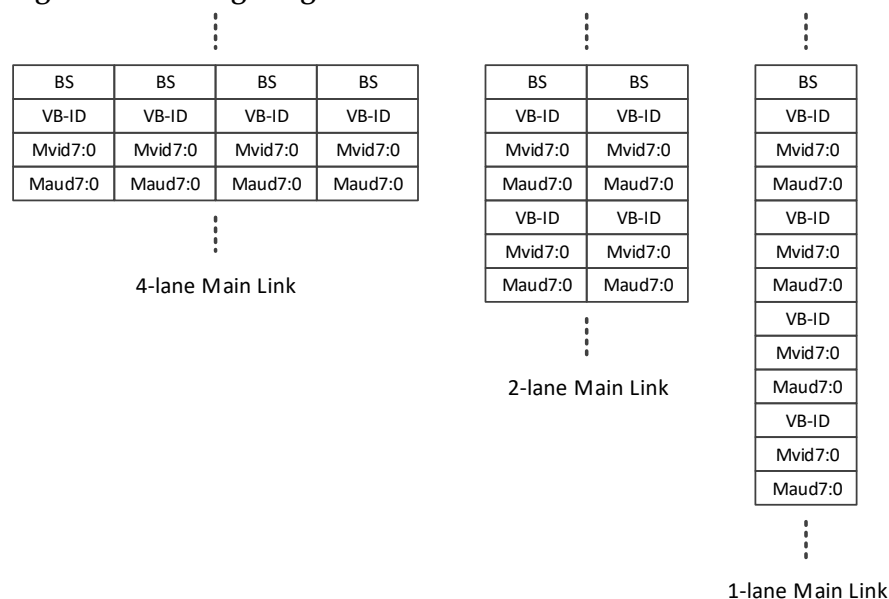
The video stream data packet format is shown in Figure 3-4.

Figure 3-4 Diagram of Video Stream Data Packet Format

The BS of each lane must be followed by the VB-ID, Mvid7:0 and Maud7:0.

Regardless of the number of lanes, VB-ID, Mvid7:0, and Maud7:0 must be transmitted four times, as shown in Figure 3-5.

Figure 3-5 Packing Diagram for VB-ID and Others



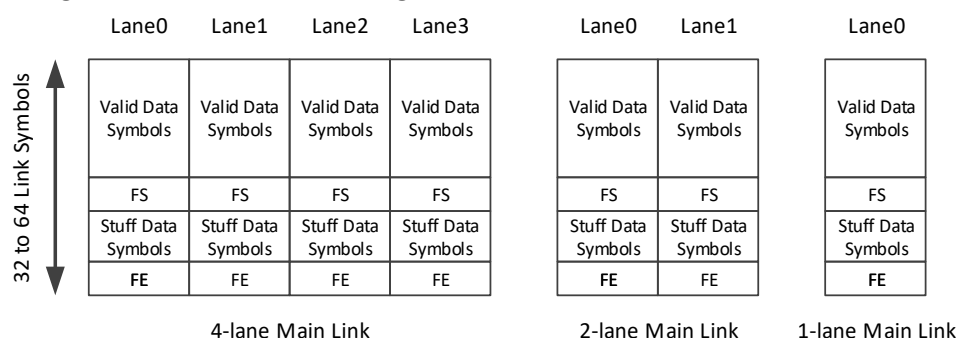
To prevent link bandwidth overload, the packed data rates must be less than or equal to the link symbol rates. When the packed data rate is lower than the link symbol rate, the link layer must perform symbol stuffing.

Symbol stuffing consists of stuffing frame symbols and dummy data symbols. Stuffing frame symbols consist of FS and FE, and are located within each Transfer Unit (TU). Dummy data symbols, placed between FS and FE, must be 0x00 before scrambling, as shown in Figure 3-6.

The size of a Transfer Unit must range from 32 to 64 symbols per lane.

The last TU in each line of video data does not require FS and FE stuffing.

Figure 3-6 Transfer Unit Diagram



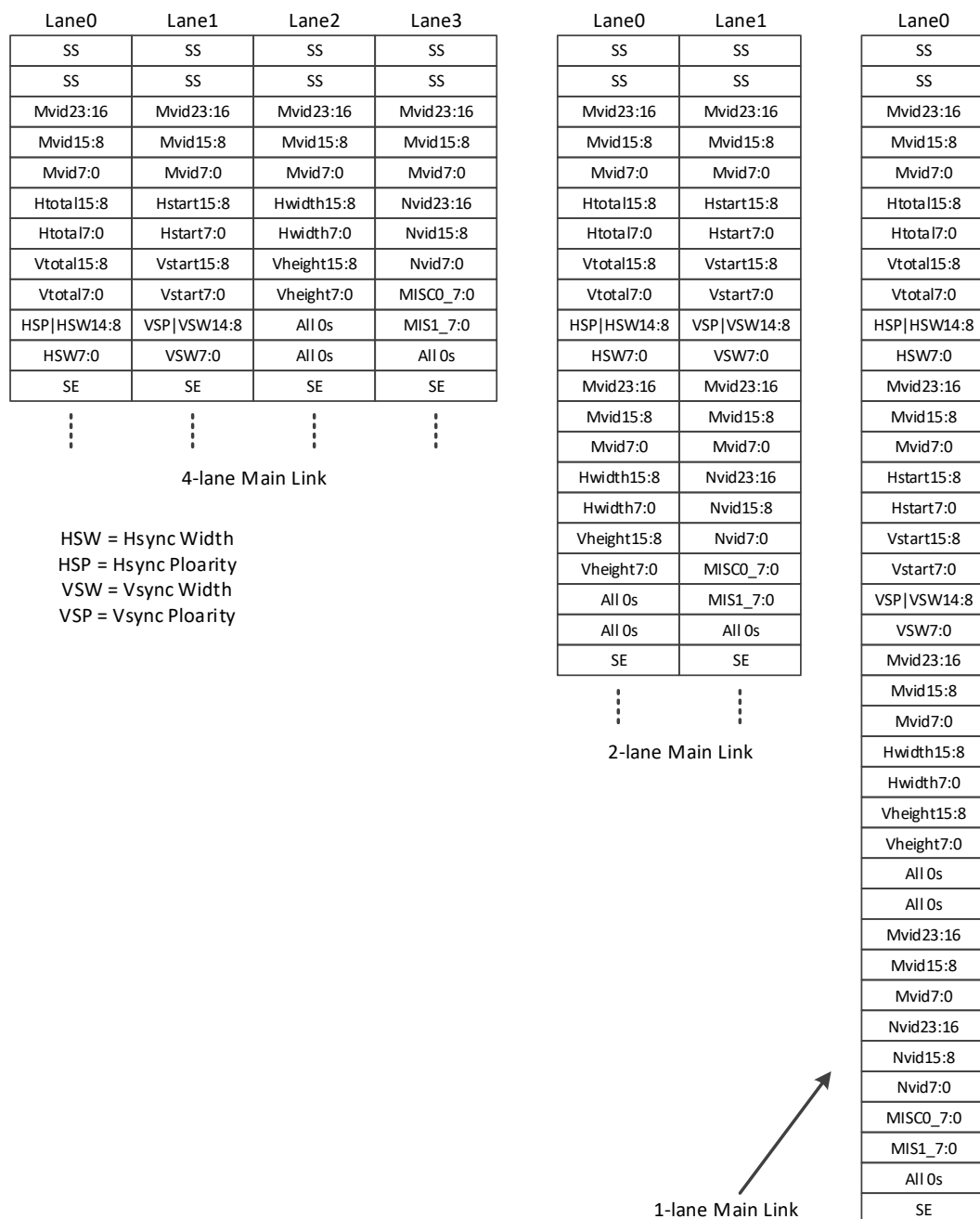
The Main Stream Attribute (MSA) data is transmitted once per frame during the vertical blanking period of the main video stream. The attribute data includes the following:

1. M and N values for video stream clock recovery (24 bits each)
2. Horizontal total pixels and vertical total lines (16 bits each)

3. Horizontal active start pixels and vertical active start lines (16 bits each)
4. Horizontal and vertical sync signal polarity and width (1 bit for polarity and 15 bits for width)
5. Horizontal active pixels and vertical active lines (16 bits each)
6. Miscellaneous 0 (MISC0, 8 bits)
7. Miscellaneous 1 (MISC1, 8 bits)

To distinguish the Main Stream Attribute data packet from the second data packet, two consecutive "SS" symbols are sent through each lane, as shown in Figure 3-7.

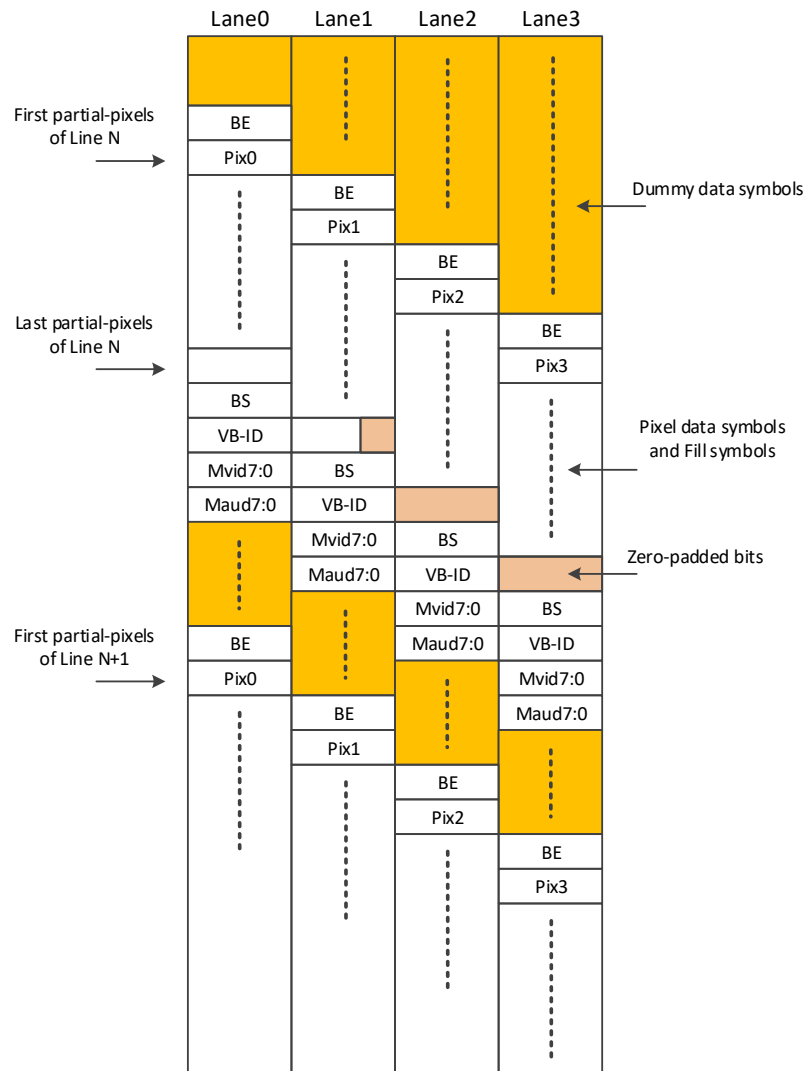
Figure 3-7 MSA Transmission Diagram



Skewing and Deskewing Modules

Adjacent lanes must insert two Link Clock (LS_Clk) skew cycles to improve the link's resistance to external noise, as shown in Figure 3-8.

Figure 3-8 Lane Skewing Diagram



Scrambler and Descrambler Modules

To reduce EMI, scrambling is performed before 8B/10B encoding. The 16-bit LFSR polynomial is $G(X) = X^{16} + X^5 + X^4 + X^3 + 1$. Each byte of data is scrambled/descrambled with the most significant 8 bits of the LFSR in reverse bit order.

$$\{D'[7], D'[6], D'[5], D'[4], D'[3], D'[2], D'[1], D'[0]\} = \{D[7], D[6], D[5], D[4], D[3], D[2], D[1], D[0]\} \wedge \{LFSR[8], LFSR[9], LFSR[10], LFSR[11], LFSR[12], LFSR[13], LFSR[14], LFSR[15]\}$$

The SR symbol or SR BF BF SR symbol sequence is used to reset the LFSR to the initial value FFFFh (or the initial value is FFFEh in the eDP standard).

3.3.2 Sideband Control

Hot-Plug Detect

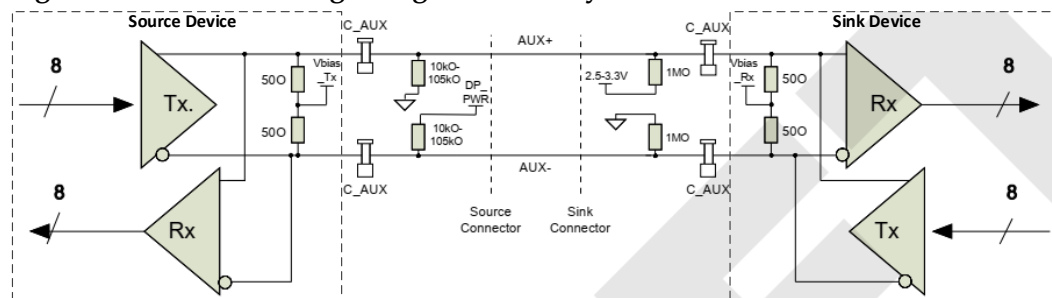
The Hot-Plug Detect signal is an interrupt request signal sent from the sink device to the source device. It is a unidirectional signal, active-high. When the source detects this signal as high, it indicates that a sink device is connected. If it is low, it indicates no sink device is connected. The HPD signal IO level standard is 3.3V.

Auxiliary Channel

The Auxiliary Channel is an independent bidirectional auxiliary channel in the DP interface. It uses AC-coupled differential transmission and is a bidirectional half-duplex channel with a rate of 1Mbps. Manchester II encoding is used to transmit control commands. The recommended IO level standard for the auxiliary channel is 1.8V.

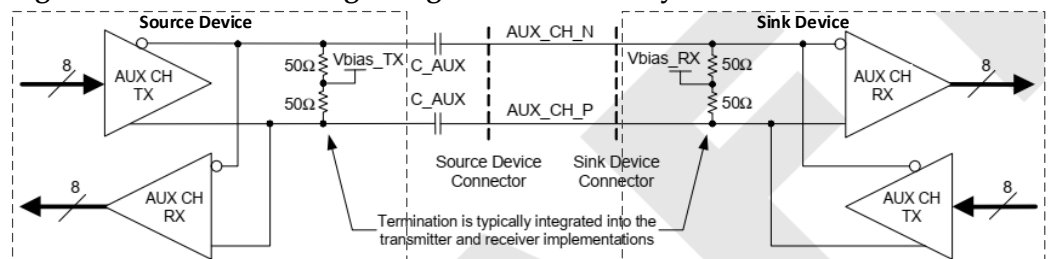
If it is used as a DP interface connected via cable, the external circuit is recommended as shown in Figure 3-9.

Figure 3-9 Differential Signaling for Auxiliary Channel



If it is used as an eDP interface for on-board connection, the external circuit is recommended as shown in Figure 3-10.

Figure 3-10 Differential Signaling for eDP Auxiliary Channel



3.4 Port List

3.4.1 Gowin EDP Encoder IP Ports

The IO port details of Gowin EDP Encoder IP are shown in Table 3-4. The IO ports vary slightly depending on the parameters.

Table 3-4 I/O List of Gowin EDP Encoder IP

Signal Name	I/O	Description	Note
I_rst_n	I	Reset signal, active-low	The I/O of all the signals take EDP Encode IP as a reference.
I_tx_tps1_en	I	TPS1 training data enable, and it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
I_tx_tps2_en	I	TPS2 training data enable, and it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	
I_tx_tps3_en	I	TPS3 training data enable, and it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
I_tx_tps4_en	I	TPS4 training data enable, and it is valid when the parameter "Training Pattern TPS4 Enable" is checked.	
I_tx_video_data_en	I	TX video data enable	
I_frame_type	I	Frame type control signal 0: Default frame 1: Enhanced frame	
I_scrm_value	I	Scrambler reset control signal 0: Reset to the initial value 0xFFFFF 1: Reset to the initial value 0xFFFFE	
I_vb_id	I	VB-ID parameter	
I_msa_mvid	I	Mvid parameter in MSA	
I_msa_nvid	I	Nvid parameter in MSA	
I_tu_size	I	TU size parameter, with range of 32~64. - When User Clock Ratio is 1:20, it must be a multiple of 2. - When User Clock Ratio is 1:40, it must be a multiple of 4. - When User Clock Ratio is 1:80, it must be a multiple of 8.	
I_msa_htotal	I	Parameter of horizontal total pixels in MSA	
I_msa_vtotal	I	Parameter of vertical total pixels in MSA	
I_msa_hstart	I	Parameter of horizontal active start pixels in MSA	
I_msa_vstart	I	Parameter of vertical active start lines in MSA	
I_msa_hwidth	I	Parameter of horizontal active pixels in MSA	
I_msa_vheight	I	Parameter of vertical active lines in MSA	
I_msa_hspshw	I	Parameter of horizontal sync signal polarity and width in MSA.	

Signal Name	I/O	Description	Note
		[15]: Polarity of the sync signal; 0: positive polarity pulse; 1: negative polarity pulse. [14:0]: The sync signal width.	
I_msa_vspvsw	I	Parameter of vertical sync signal polarity and width in MSA. [15]: Polarity of the sync signal; 0: positive polarity pulse; 1: negative polarity pulse. [14:0]: The sync signal width.	
I_msa_misc0	I	MISC0 parameter in MSA [0]: 0 indicates the link clock and the video stream clock are asynchronous; 1 indicates the link clock and the video stream clock are synchronous.	
I_msa_misc1	I	MISC1 parameter in MSA	
I_strm_clk	I	Input video stream clock	
I_pix_vs	I	Input video stream vertical sync signal	
I_pix_de	I	Input video stream data valid signal	
I_ls_clk	I	Input link clock	
I_pix_data0	I	Input video stream 0; when the "Bits Per Pixel Coefficient" is set to 1, 2, or 4, the corresponding data formats are as follows: {B,G,R}/{B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }/{ B ₃ ,G ₃ ,R ₃ ,B ₂ ,G ₂ ,R ₂ , B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }.	
I_pix_data1	I	Input video stream 1, and it is supported only when the link is configured with 2 or 4 lanes. In this case, when the "Bits Per Pixel Coefficient" is set to 1, 2, or 4, the corresponding data formats are as follows: {B,G,R}/{B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }/{ B ₃ ,G ₃ ,R ₃ ,B ₂ ,G ₂ ,R ₂ , B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }.	
I_pix_data2	I	Input video stream 2, and it is supported only when the link is configured with 4 lanes. In this case, when the "Bits Per Pixel Coefficient" is set to 1, 2, or 4, the corresponding data formats are as follows: {B,G,R}/{B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }/{ B ₃ ,G ₃ ,R ₃ ,B ₂ ,G ₂ ,R ₂ , B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }.	
I_pix_data3	I	Input video stream 3, and it is supported only when the link is configured with 4 lanes. In this case, when the "Bits Per Pixel Coefficient" is set to 1, 2, or 4, the corresponding data formats are as follows:{B,G,R}/{B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }/{ B ₃ ,G ₃ ,R ₃ , B ₂ ,G ₂ ,R ₂ , B ₁ ,G ₁ ,R ₁ ,B ₀ ,G ₀ ,R ₀ }.	
O_tx_data_0	O	Output link data 0 - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is	

Signal Name	I/O	Description	Note
		64 bits.	
O_tx_data_1	O	Output link data 1, and it is supported when the link is 2/4 lanes. - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
O_tx_data_2	O	Output link data 2, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
O_tx_data_3	O	Output link data 3, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
O_k_code_0	O	Output link K character flag 0 - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
O_k_code_1	O	Output link K character flag 1, and it is supported when the link is 2/4 lanes. - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
O_k_code_2	O	Output link K character flag 2, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
O_k_code_3	O	Output link K character flag 3, and it is supported when the link is 4 lanes.	

Signal Name	I/O	Description	Note
		- When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
O_data_bf_scram_0	O	The link data 0 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 16 bits. - When the User Clock Ratio is 1:40, the data width is 32 bits. - When the User Clock Ratio is 1:80, the data width is 64 bits.	
O_data_bf_scram_1	O	The link data 1 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 16 bits. - When the User Clock Ratio is 1:40, the data width is 32 bits. - When the User Clock Ratio is 1:80, the data width is 64 bits.	
O_data_bf_scram_2	O	The link data 2 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 16 bits. - When the User Clock Ratio is 1:40, the data width is 32 bits. - When the User Clock Ratio is 1:80, the data width is 64 bits.	
O_data_bf_scram_3	O	The link data 3 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 16 bits. - When the User Clock Ratio is 1:40, the data width is 32 bits. - When the User Clock Ratio is 1:80, the data width is 64 bits.	
O_kcode_bf_scram_0	O	The K-character flag 0 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 2 bits. - When the User Clock Ratio is 1:40, the data width is 4 bits. - When the User Clock Ratio is 1:80, the data width is 8 bits.	

Signal Name	I/O	Description	Note
O_kcode_bf_scram_1	O	The K-character flag 1 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 2 bits. - When the User Clock Ratio is 1:40, the data width is 4 bits. - When the User Clock Ratio is 1:80, the data width is 8 bits.	
O_kcode_bf_scram_2	O	The K-character flag 2 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 2 bits. - When the User Clock Ratio is 1:40, the data width is 4 bits. - When the User Clock Ratio is 1:80, the data width is 8 bits.	
O_kcode_bf_scram_3	O	The K-character flag 3 before scrambling, and it is supported when "TX Debug Enable" is set to Yes. - When the User Clock Ratio is 1:20, the data width is 2 bits. - When the User Clock Ratio is 1:40, the data width is 4 bits. - When the User Clock Ratio is 1:80, the data width is 8 bits.	

Note!

- When the Lanes Number is set to 1 lane, the number of input pixels is 1, 2, or 4 when the Bits Per Pixel Coefficient is 1, 2, or 4, respectively.
- When the Lanes Number is set to 2 lanes, the number of input pixels is 2, 4, or 8 when the Bits Per Pixel Coefficient is 1, 2, or 4, respectively.
- When the Lanes Number is set to 4 lanes, the number of input pixels is 4, 8, or 16 when the Bits Per Pixel Coefficient is 1, 2, or 4, respectively.

3.4.2 Gowin EDP Decoder IP Ports

The IO port details of Gowin EDP Decoder IP are shown in Table 3-5. The IO ports vary slightly depending on the parameters.

Table 3-5 I/O List of Gowin EDP Decoder IP

Signal Name	I/O	Description	Note
I_rst_n	I	Reset signal, active-low	The I/O of all the signals take EDP Decoder IP as a reference.
I_ls_clk	I	Input link clock	
I_tps1_en	I	TPS1 training data enable, and it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
I_tps2_en	I	TPS2 training data enable, and it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	
I_tps3_en	I	TPS3 training data enable, and it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
I_frame_type	I	Frame type control signal 0: Default frame 1: Enhanced frame	
I_scrm_value	I	Scrambler reset control signal 0: Reset to the initial value 0xFFFF 1: Reset to the initial value 0xFFFE	
O_vb_id	O	VB-ID parameter	
O_mvid70	O	Low 8 bits of Mvid parameter in MSA	
O_maud70	O	Low 8 bits of audio Maud parameter	
O_mvid	O	Mvid parameter in MSA	
O_nvid	O	Nvid parameter in MSA	
O_htotal	O	Parameter of horizontal total pixels in MSA	
O_vtotal	O	Parameter of vertical total pixels in MSA	
O_hstart	O	Parameter of horizontal active start pixels in MSA	
O_vstart	O	Parameter of vertical active start lines in MSA	
O_hwidth	O	Parameter of horizontal active pixels in MSA	
O_vheight	O	Parameter of vertical active lines in MSA	
O_hsw	O	Parameter of horizontal sync signal width in MSA	
O_vsw	O	Parameter of vertical sync signal width in MSA	
O_hsp	O	Parameter of horizontal sync signal polarity in MSA	
O_vsp	O	Parameter of vertical sync signal polarity in MSA	
O_misc0	O	MISC0 parameter in MSA	
O_misc1	O	MISC1 parameter in MSA	
O_ls_vs	O	Vertical sync signal for the link clock domain	
O_ls_hs	O	Horizontal sync signal for the link clock domain	
I_kcode0	I	Input lane0 link K character flag 0	

Signal Name	I/O	Description	Note
		- When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
I_data0	I	Input lane0 link data 0 - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
I_kcode1	I	Input lane1 link K character flag 1, and it is supported when the link is 2/4 lanes. - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
I_data1	I	Input lane1 link data 1, and it is supported when the link is 2/4 lanes. - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
I_kcode2	I	Input lane2 link K character flag 2, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is 4 bits. - When User Clock Ratio is 1:80, the width is 8 bits.	
I_data2	I	Input lane2 link data 2, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 16 bits. - When User Clock Ratio is 1:40, the width is 32 bits. - When User Clock Ratio is 1:80, the width is 64 bits.	
I_kcode3	I	Input lane3 link K character flag 3, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the width is 2 bits. - When User Clock Ratio is 1:40, the width is	

Signal Name	I/O	Description	Note
		4 bits. ● When User Clock Ratio is 1:80, the width is 8 bits.	
I_data3	I	Input lane3 link data 3, and it is supported when the link is 4 lanes. ● When User Clock Ratio is 1:20, the width is 16 bits. ● When User Clock Ratio is 1:40, the width is 32 bits. ● When User Clock Ratio is 1:80, the width is 64 bits.	
O_rx0_tps1_ok	O	Lane 0 TPS1 Data training success flag, and it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
O_rx0_tps2_ok	O	Lane 0 TPS2 Data training success flag, and it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	
O_rx0_tps3_ok	O	Lane 0 TPS3 Data training success flag, and it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
O_rx1_tps1_ok	O	Lane1 TPS1 data training success flag, and it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
O_rx1_tps2_ok	O	Lane1 TPS2 data training success flag, and it is supported when the link is 2/4 lanes; it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	
O_rx1_tps3_ok	O	Lane1 TPS3 data training success flag, and it is supported when the link is 2/4 lanes; it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
O_rx2_tps1_ok	O	Lane2 TPS1 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
O_rx2_tps2_ok	O	Lane2 TPS2 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	
O_rx2_tps3_ok	O	Lane2 TPS3 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
O_rx3_tps1_ok	O	Lane3 TPS1 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS1 Enable" is checked.	
O_rx3_tps2_ok	O	Lane3 TPS2 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS2 Enable" is checked.	

Signal Name	I/O	Description	Note
O_rx3_tps3_ok	O	Lane3 TPS3 data training success flag, and it is supported when the link is 4 lanes; it is valid when the parameter "Training Pattern TPS3 Enable" is checked.	
O_unp0_data	O	Lane 0 unpacked data - When User Clock Ratio is 1:20, the format is {B,G,R}. - When User Clock Ratio is 1:40, the format is {B₁,G₁,R₁,B₀,G₀,R₀}. - When User Clock Ratio is 1:80, the format is {B₃,G₃,R₃,B₂,G₂,R₂, B₁,G₁,R₁,B₀,G₀,R₀}.	
O_unp0_data_valid	O	Lane 0 unpacked data valid signal	
O_unp1_data	O	Lane1 unpacked data, and it is supported when the link is 2/4 lanes. - When User Clock Ratio is 1:20, the format is {B,G,R}. - When User Clock Ratio is 1:40, the format is {B₁,G₁,R₁,B₀,G₀,R₀}. - When User Clock Ratio is 1:80, the format is {B₃,G₃,R₃,B₂,G₂,R₂, B₁,G₁,R₁,B₀,G₀,R₀}.	
O_unp1_data_valid	O	Output lane1 unpacked data valid signal, and it is supported when the link is 2/4 lanes	
O_unp2_data	O	Lane2 unpacked data, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the format is {B,G,R}. - When User Clock Ratio is 1:40, the format is {B₁,G₁,R₁,B₀,G₀,R₀}. - When User Clock Ratio is 1:80, the format is {B₃,G₃,R₃,B₂,G₂,R₂, B₁,G₁,R₁,B₀,G₀,R₀}.	
O_unp2_data_valid	O	Lane2 unpacked data valid signal, and it is supported when the link is 4 lanes	
O_unp3_data	O	Lane3 unpacked data, and it is supported when the link is 4 lanes. - When User Clock Ratio is 1:20, the format is {B,G,R}. - When User Clock Ratio is 1:40, the format is {B₁,G₁,R₁,B₀,G₀,R₀}. - When User Clock Ratio is 1:80, the format is {B₃,G₃,R₃,B₂,G₂,R₂, B₁,G₁,R₁,B₀,G₀,R₀}.	
O_unp3_data_valid	O	Lane3 unpacked data valid signal, and it is supported when the link is 4 lanes.	
O_data0_af_descr	O	Output the link data of lane 0 after descrambling, and it is supported when "RX Debug Enable" is set to "Yes". - When the User Clock Ratio is 1:20, the data width is 16 bits. - When the User Clock Ratio is 1:40, the data width is 32 bits.	

Signal Name	I/O	Description	Note
		When the User Clock Ratio is 1:80, the data width is 64 bits.	
O_data1_af_descr	O	Output the link data of lane 1 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 2 or 4 lanes. - When User Clock Ratio is 1:20, the data width is 16 bits. - When User Clock Ratio is 1:40, the data width is 32 bits. - When User Clock Ratio is 1:80, the data width is 64 bits.	
O_data2_af_descr	O	Output the link data of lane 2 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 4 lanes. - When User Clock Ratio is 1:20, the data width is 16 bits. - When User Clock Ratio is 1:40, the data width is 32 bits. - When User Clock Ratio is 1:80, the data width is 64 bits.	
O_data3_af_descr	O	Output the link data of lane 3 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 4 lanes. - When User Clock Ratio is 1:20, the data width is 16 bits. - When User Clock Ratio is 1:40, the data width is 32 bits. - When User Clock Ratio is 1:80, the data width is 64 bits.	
O_kcode0_af_descr	O	Output the K-character flag of lane 0 after descrambling, and it is supported when "RX Debug Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - when User Clock Ratio is 1:40, the bit width is 4 bits. - when User Clock Ratio is 1:80, the bit width is 8 bits.	
O_kcode 1_af_descr	O	Output the K-character flag of lane 1 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 2 or 4 lanes. - When User Clock Ratio is 1:20, the bit width is 2 bits. - when User Clock Ratio is 1:40, the bit width is 4 bits. - when User Clock Ratio is 1:80, the bit width is 8 bits.	
O_kcode 2_af_descr	O	Output the K-character flag of lane 2 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 4	

Signal Name	I/O	Description	Note
		lanes. - When User Clock Ratio is 1:20, the bit width is 2 bits. - when User Clock Ratio is 1:40, the bit width is 4 bits. - when User Clock Ratio is 1:80, the bit width is 8 bits.	
O_kcode 3_af_descr	O	Output the K-character flag of lane 2 after descrambling; it is supported when "RX Debug Enable" is set to "Yes" and the link is set to 4 lanes. - When User Clock Ratio is 1:20, the bit width is 2 bits. - when User Clock Ratio is 1:40, the bit width is 4 bits. - when User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp0_kcode	O	Output the K-character flag of the second data packet on lane 0; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp0_data	O	Output the link data of the second data packet on lane 0; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_sdp0_vaild	O	Output the data valid flag of the second data packet on lane 0; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp1_kcode	O	Output the K-character flag of the second data packet on lane 1; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits.	

Signal Name	I/O	Description	Note
		When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp1_data	O	Output the link data of the second data packet on lane 1; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". When User Clock Ratio is 1:20, the bit width is 16 bits. When User Clock Ratio is 1:40, the bit width is 32 bits. When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_sdp1_vaild	O	Output the data valid flag of the second data packet on lane 1; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp2_kcode	O	Output the K-character flag of the second data packet on lane 2; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp2_data	O	Output the link data of the second data packet on lane 2; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_sdp2_vaild	O	Output the data valid flag of the second data packet on lane 2; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp3_kcode	O	Output the K-character flag of the second data packet on lane 3; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". When User Clock Ratio is 1:20, the bit	

Signal Name	I/O	Description	Note
		width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_sdp3_data	O	Output the link data of the second data packet on lane 3; it is supported when RX Secondary Data Packet Enable is set to Yes." - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_sdp3_vaild	O	Output the data valid flag of the second data packet on lane 3; it is supported when "RX Secondary Data Packet Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_hdcp_descr0_kcode	O	Output the K-character flag of the HDCP data on lane 0; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_hdcp_descr0_data	O	Output the HDCP link data on lane 0; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_hdcp_descr1_kcode	O	Output the K-character flag of the HDCP data on lane 1; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_hdcp_descr1_data	O	Output the HDCP link data on lane 1; it is	

Signal Name	I/O	Description	Note
		supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_hdcp_descr2_kcode	O	Output the K-character flag of the HDCP data on lane 2; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_hdcp_descr2_data	O	Output the HDCP link data on lane 2; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
O_hdcp_descr3_kcode	O	Output the K-character flag of the HDCP data on lane 3; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
O_hdcp_descr3_data	O	Output the HDCP link data on lane 3; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
I_hdcp_descr0_kcode	I	Input the K-character flag of the HDCP data on lane 0; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits.	

Signal Name	I/O	Description	Note
		When User Clock Ratio is 1:80, the bit width is 8 bits.	
I_hdcp_descr0_data	I	Input the HDCP link data on lane 0; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
I_hdcp_descr1_kcode	I	Input the K-character flag of the HDCP data on lane 1; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
I_hdcp_descr1_data	I	Input the HDCP link data on lane 1; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
I_hdcp_descr2_kcode	I	Input the K-character flag of the HDCP data on lane 2; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
I_hdcp_descr2_data	I	Input the HDCP link data on lane 2; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
I_hdcp_descr3_kcode	I	Input the HDCP data K-character flag on lane 3; it is supported when "RX HDCP Enable" is set to "Yes". When User Clock Ratio is 1:20, the bit	

Signal Name	I/O	Description	Note
		width is 2 bits. - When User Clock Ratio is 1:40, the bit width is 4 bits. - When User Clock Ratio is 1:80, the bit width is 8 bits.	
I_hdcp_descr3_data	I	Input the HDCP lane link data on lane 3; it is supported when "RX HDCP Enable" is set to "Yes". - When User Clock Ratio is 1:20, the bit width is 16 bits. - When User Clock Ratio is 1:40, the bit width is 32 bits. - When User Clock Ratio is 1:80, the bit width is 64 bits.	
I_lan_num	I	- The lane number input to the Decode module specifies the actual number of lanes used for decoding by the Decode module and supports backward compatibility with different lane count configurations. The value is provided by the AUX RX module and passed to the Decode module. - For 4-lane mode, the configurable values are 4, 2, and 1. - For 2-lane mode, the configurable values are 2 and 1. - For 1-lane mode, the configurable value is 1.	

Note!

- When the User Clock Ratio parameter is set to 1:20, each lane outputs single pixel data. When the User Clock Ratio parameter is set to 1:40, each lane outputs two-pixel data. When the User Clock Ratio parameter is set to 1:80, each lane outputs four-pixel data.
- The output unpacked data valid signal (unp_data_valid) is a non-continuous signal per line, and it is different from the DE signal.

3.5 Parameter Configuration

3.5.1 EDP Encoder Parameters

Table 3-6 EDP Encoder Parameters

Name	Range	Default	Description
User Clock Ratio	1:20/1:40/1:80	1:20	Ratio of user clock to serial clock
Lanes Number	1/2/4	1	Number of lanes for main link
Bits Per Pixel	18/24/30	24	Bits per pixel
Bits Per Pixel Coefficient	1/2/4	1	Coefficient of bits per pixel
TX Line Buffer Depth	64/128/256	64	TX line buffer depth
Mvid Customer	Checked/Unchecked	Checked	User-controlled enable via I_msa_mvid
Training Pattern TPS1 Enable	Checked/Unchecked	Checked	Training Pattern TPS1 enable

Name	Range	Default	Description
Training Pattern TPS2 Enable	Checked/Unchecked	Checked	Training Pattern TPS2 enable
Training Pattern TPS3 Enable	Checked/Unchecked	Unchecked	Training Pattern TPS3 enable
Training Pattern TPS4 Enable	Checked/Unchecked	Unchecked	Training Pattern TPS4 enable
TX Debug Enable	Checked/Unchecked	Unchecked	Enable control for outputting pre-scrambled link data and K-character flag.

3.5.2 EDP Decoder Parameters

Table 3-7 EDP Decoder Parameters

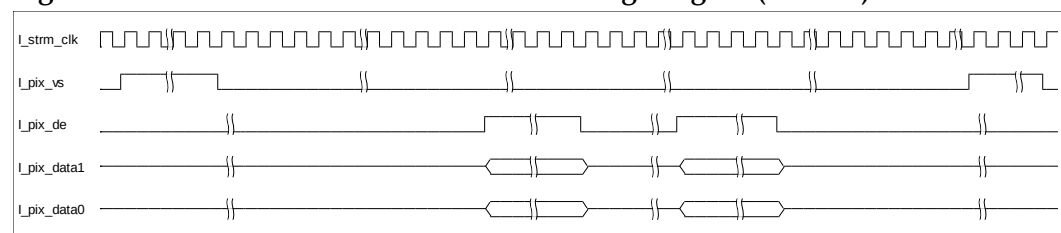
Name	Range	Default	Description
User Clock Ratio	1:20/1:40/1:80	1:20	Ratio of user clock to serial clock
Lanes Number	1/2/4	1	Number of lanes for main link
Bits Per Pixel	24	24	Bits per pixel
RX Deskew Enable	Checked/Unchecked	Checked	RX deskew enable; If the EDP PHY has the RX Channel Bonding enabled, you can disable RX deskew module.
Training Pattern TPS1 Enable	Checked/Unchecked	Checked	Training Pattern TPS1 enable
Training Pattern TPS2 Enable	Checked/Unchecked	Checked	Training Pattern TPS2 enable
Training Pattern TPS3 Enable	Checked/Unchecked	Unchecked	Training Pattern TPS3 enable
RX Debug Enable	Checked/Unchecked	Unchecked	Enable control for outputting descrambled link data and K-character flag.
RX HDCP Enable	Checked/Unchecked	Unchecked	HDCP parsing enable
RX Secondary Data Packet Enable	Checked/Unchecked	Unchecked	Secondary data packet output enable

3.6 Timing Description

This section describes the timing of Gowin EDP Encoder Decoder IP.

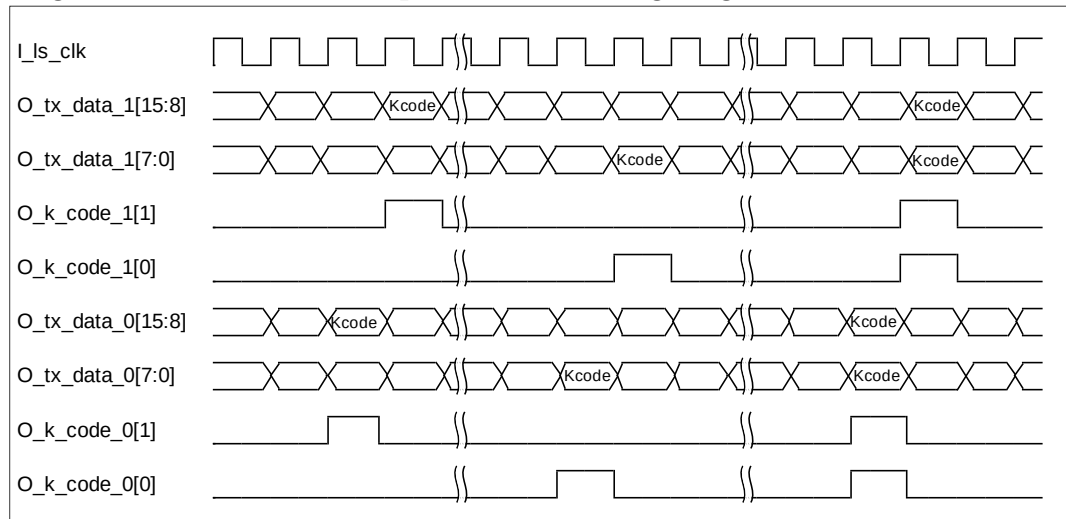
The EDP Encoder video interface timing (2 lanes) is as shown in Figure 3-11.

Figure 3-11 EDP Encoder Video Interface Timing Diagram(2 Lanes)

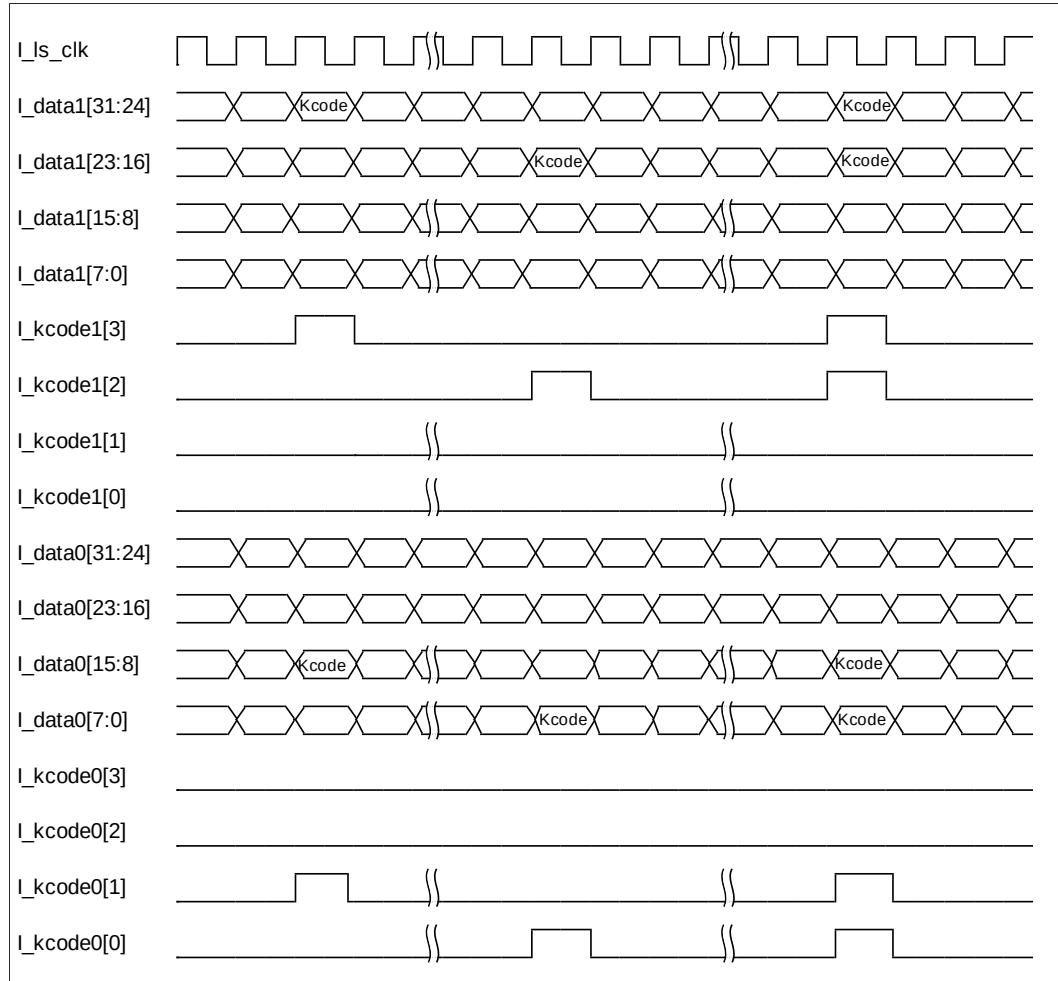


The EDP Encoder output interface timing (2 lanes) with the data width 16 bits is as shown in Figure 3-12.

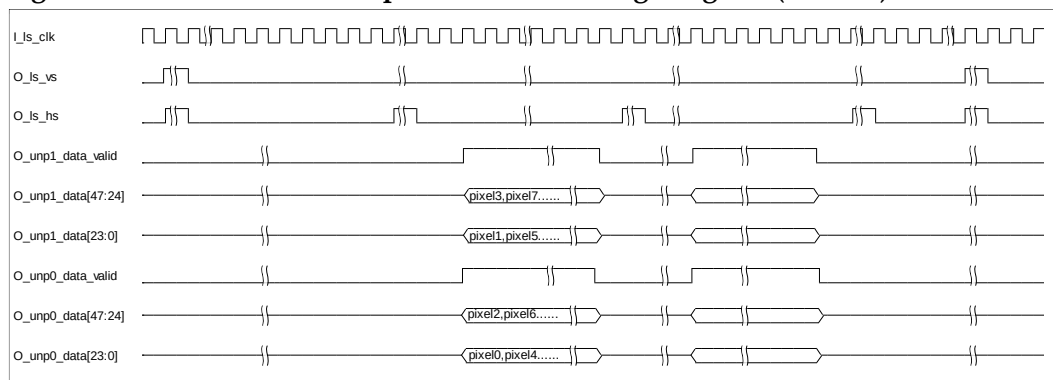
Figure 3-12 EDP Encoder Output Interface Timing Diagram (2 Lanes)



The EDP Decoder input interface timing (2 lanes) with data width 32 bits is as shown in Figure 3-13.

Figure 3-13 EDP Decoder Input Interface Timing Diagram (2 Lanes)

The EDP Decoder output interface timing (2 lanes) with data width 48 bits is as shown in Figure 3-14.

Figure 3-14 EDP Decoder Output Interface Timing Diagram (2 Lanes)**Note!**

When the User Clock Ratio is 1:40 and the Lanes Number is 2 or 4 lanes, 48 bits of pixel data per lane are output non-sequentially in the order, as shown in above figure.

3.7 EDP Encoder Port Parameter Calculation

This manual illustrates the port parameter calculation method with three examples.

- **f_Pixel_Clk**: Video stream clock frequency. For parameter calculation, it must be based on the VESA single-pixel mode clock frequency. The actual video stream clock frequency should be divisible by the number of pixels per clock.
- **f_link_Clk**: Link clock frequency. For parameter calculation, it must be based on the link clock frequency with a 1:10 clock ratio. The actual link clock frequency is calculated based on the configured clock ratio.

The video timing parameters are described as follows:

- **htotal**: Total number of pixels per line
- **vtotal**: Total number of lines per frame
- **hstart**: Number of pixels from the start of hsync to the start of DE, calculated as $hstart = hsync + hbporch$
- **vstart**: Number of lines from the start of vsync to the start of DE in the first line, calculated as $vstart = vsync + vbporch$
- **hwidth**: Number of active pixels per line
- **vheight**: Number of active lines per frame
- **hsw**: Horizontal sync width, in pixels
- **vsw**: Vertical sync width, in lines
- **hsp**: Horizontal sync polarity; 0 = positive polarity, 1 = negative polarity
- **vsp**: Vertical sync polarity; 0 = positive polarity, 1 = negative polarity
- **misc0**: Master video stream miscellaneous attribute 0 (bit0: 0 = link clock and video clock are asynchronous, 1 = link clock and video clock are synchronous)
- **misc2**: Master video stream miscellaneous attribute 1

Example One

For the eDP panel with a resolution of 3840×2160@60Hz, the serial link rate is 5.4Gbps with 4 lanes. The video stream pixel clock frequency is $f_Strm_Clk = 533.280\text{MHz}$, and the link clock frequency is calculated as $f_LS_Clk = \text{Serial Link Rate}/10 = 5.4\text{Gbps}/10 = 540\text{MHz}$. Video timing parameters include $htotal = 4000$, $vtotal = 2222$, $hstart = 48$, $vstart = 6$, $hwidth = 3840$, $vheight = 2160$, $hsw = 8$, $vsw = 2$, $hsp = 0$, $vsp = 0$, $misc0 = 0x20$ and $misc1 = 0x00$, BPP (Bits Per Pixel) = 24, User Clock Ratio = 1:40.

$Nvid$ is set to 32768, and $Mvid$ is calculated using the formula $Mvid = Nvid * (f_Strm_Clk/f_LS_Clk) = 32360$.

Example Two

For the eDP panel with a resolution of 1280×800@60Hz, the serial link rate is 2.7Gbps with 2 lanes. The video stream pixel clock frequency is

$f_Strm_Clk = 72.484\text{MHz}$. The link clock frequency is calculated as $f_LS_Clk = \text{Serial Link Rate}/10 = 2.7\text{Gbps}/10 = 270\text{MHz}$. Video timing parameters include $htotal = 1452$, $vtotal = 832$, $hstart = 48$, $vstart = 6$, $hwidth = 1280$, $vheight = 800$, $hsw = 8$, $vsw = 2$, $hsp = 0$, $vsp = 0$, $misc0 = 0x20$ and $misc1 = 0x00$, BPP (Bits Per Pixel) = 24, User Clock Ratio = 1:40.

Nvid is set to 32768, and Mvid is calculated using the formula $Mvid = Nvid * (f_Strm_Clk/f_LS_Clk) = 8797$.

Example Three

For the DP monitor with a resolution of $1920 \times 1080 @ 59.954\text{Hz}$, the serial link rate is 1.62Gbps with 4 lanes. The video stream pixel clock frequency is $f_Strm_Clk = 148.386\text{MHz}$, and the link clock frequency is calculated as $f_LS_Clk = \text{Serial Link Rate}/10 = 1.62\text{Gbps}/10 = 162\text{MHz}$. Video timing parameters include $htotal=2200$, $vtotal=1125$, $hstart=192$, $vstart=41$, $hwidth=1920$, $vheight=1080$, $hsw=44$, $vsw=5$, $hsp=0$, $vsp=0$, $misc0=0x20$, $misc1=0x00$, $bpp(\text{Bits Per Pixel})=24$, User Clock Ratio=1:40.

Nvid is set to 32768, and Mvid is calculated using the formula $Mvid = Nvid * (f_Strm_Clk/f_LS_Clk) = 30014$.

4 Interface Configuration

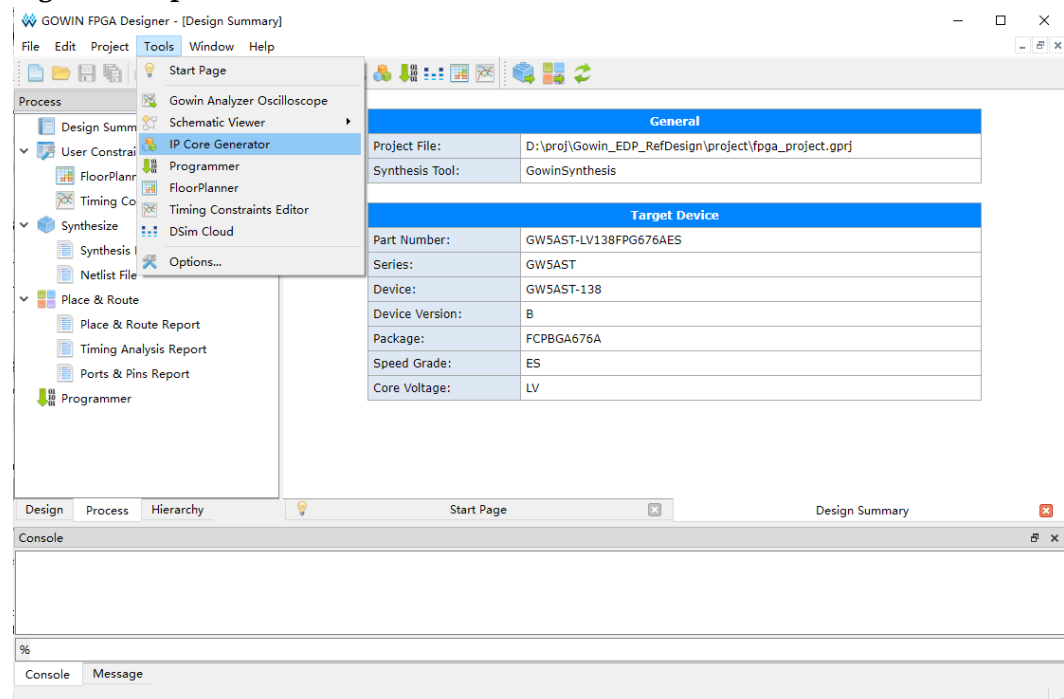
You can use IP core generator tool in Gowin Software to call and configure Gowin EDP Encoder and Decode IP.

4.1 EDP Encoder Configuration

1. Open IP Core Generator

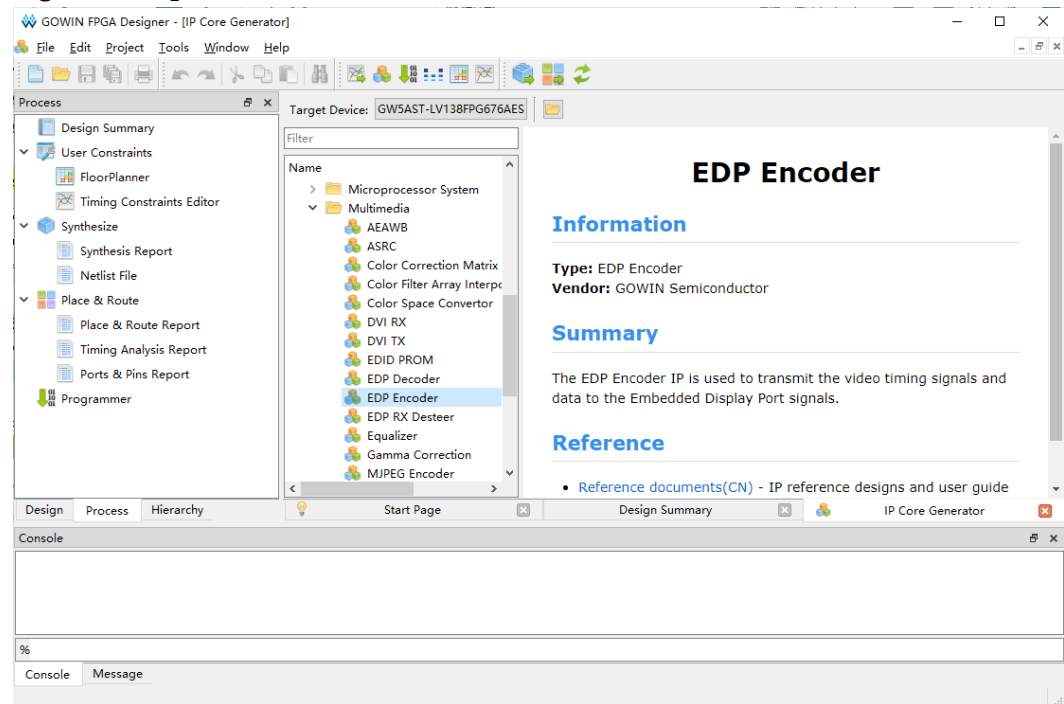
After creating the project, click the "Tools" tab in the upper left, click "IP Core Generator" from the drop-down list to open Gowin IP Core Generator, as shown in Figure 4-1.

Figure 4-1 Open IP Core Generator



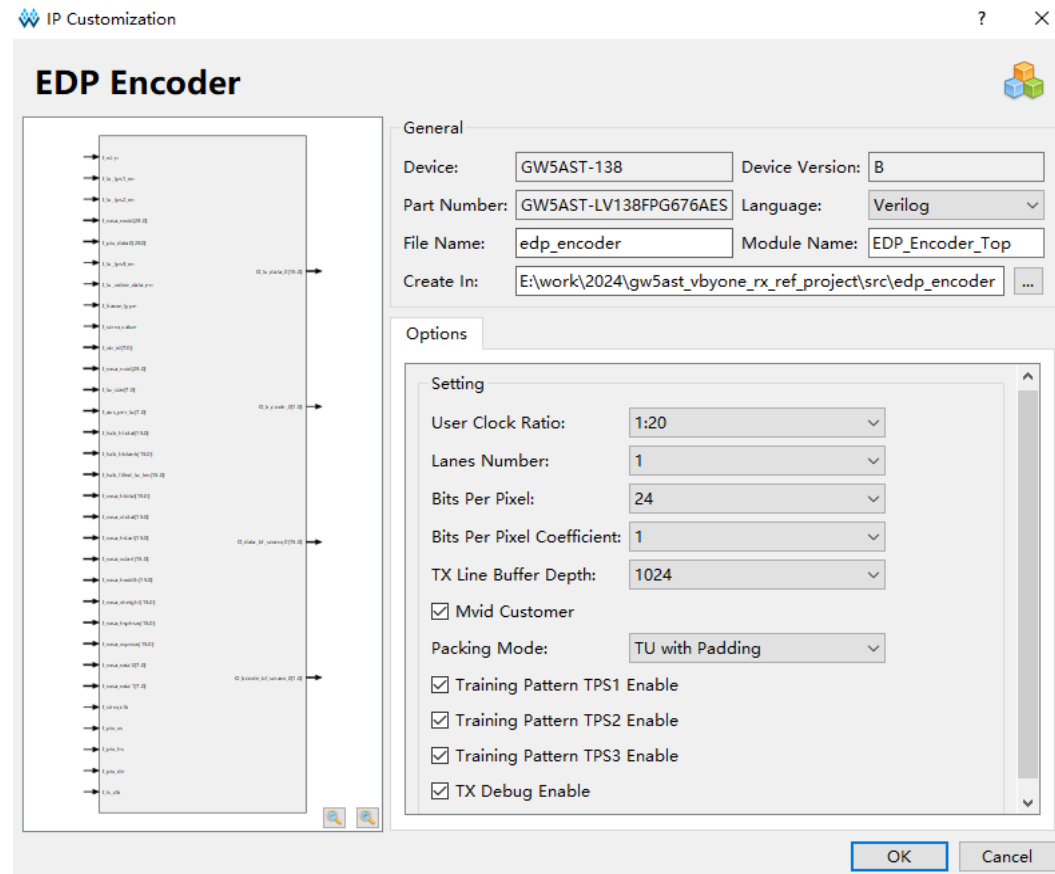
2. Open EDP Encoder IP core

Click "Multimedia" and double click "EDP Encode" to open the configuration interface, as shown in Figure 4-2.

Figure 4-2 Open EDP Encoder IP Core

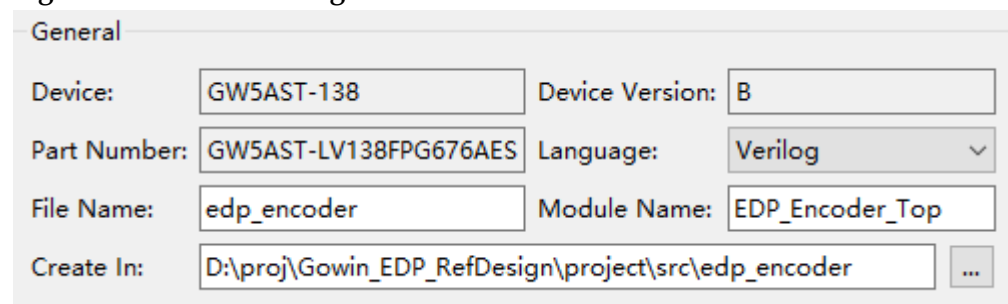
3. EDP Encoder IP Core Port Interface

On the left of the configuration interface is the port diagram of EDP Encoder IP core, as shown in Figure 4-3.

Figure 4-3 Port Diagram of EDP Encoder IP

4. Configure General Information

See the project general information in the configuration interface. Take GW5AST-138 and FPG676A package as an example. The "Module Name" displays the top-level file name of the generated project, and the default is "EDP_Encode_Top"; you can modify the name. The "File Name" displays the folder generated by the IP core, which contains the files required by EDP Encoder IP core, and the default is "edp_encode", and you can modify the path. "Create In" displays the path of IP core folder, and the default is "\project path\src\edp_encode"; you can modify the path.

Figure 4-4 General Configuration

5. Options Tab

In this tab, you can configure EDP Encoder parameters as needed.

Figure 4-5 Options Configuration

Options

Setting

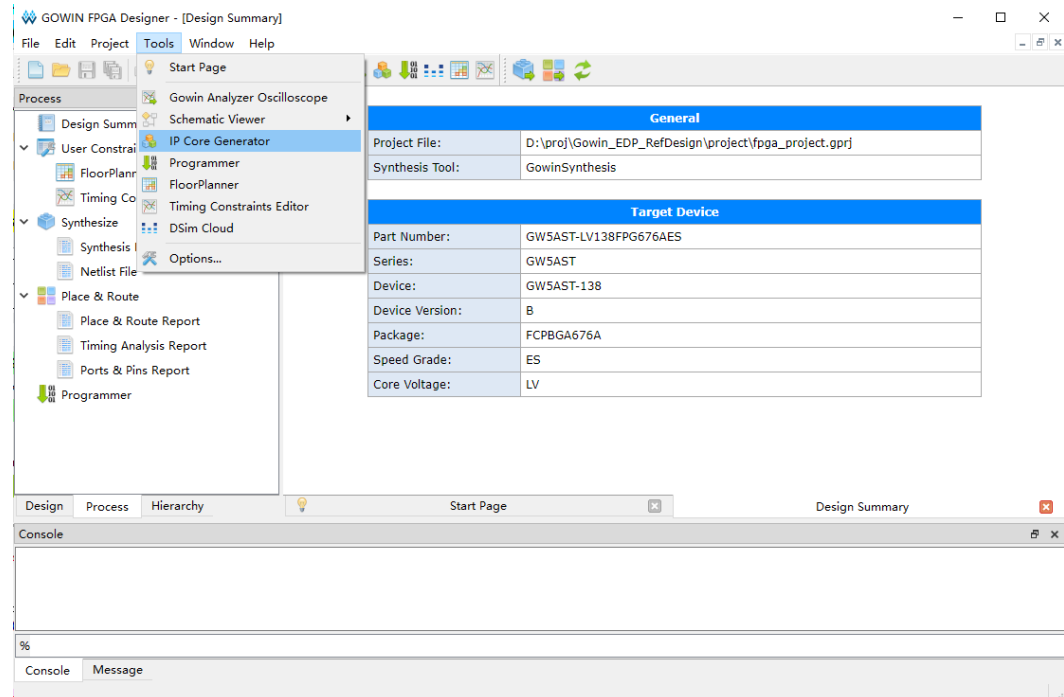
User Clock Ratio:	1:20	▼
Lanes Number:	1	▼
Bits Per Pixel:	24	▼
Bits Per Pixel Coefficient:	1	▼
TX Line Buffer Depth:	1024	▼
☒ Mvid Customer		
Packing Mode:	TU with Padding	▼
☒ Training Pattern TPS1 Enable		
☒ Training Pattern TPS2 Enable		
☐ Training Pattern TPS3 Enable		
☐ TX Debug Enable		

4.2 EDP Decoder Configuration

1. Open IP Core Generator

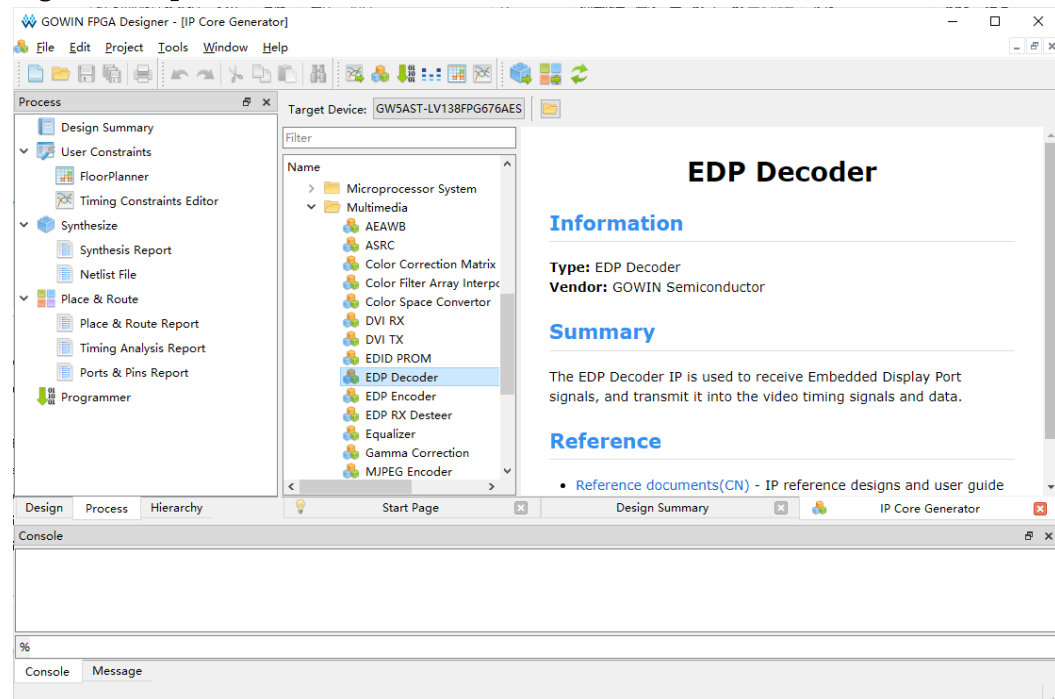
After creating the project, click the "Tools" tab in the upper left, click "IP Core Generator" from the drop-down list to open Gowin IP Core Generator, as shown in Figure 4-6.

Figure 4-6 Open IP Core Generator



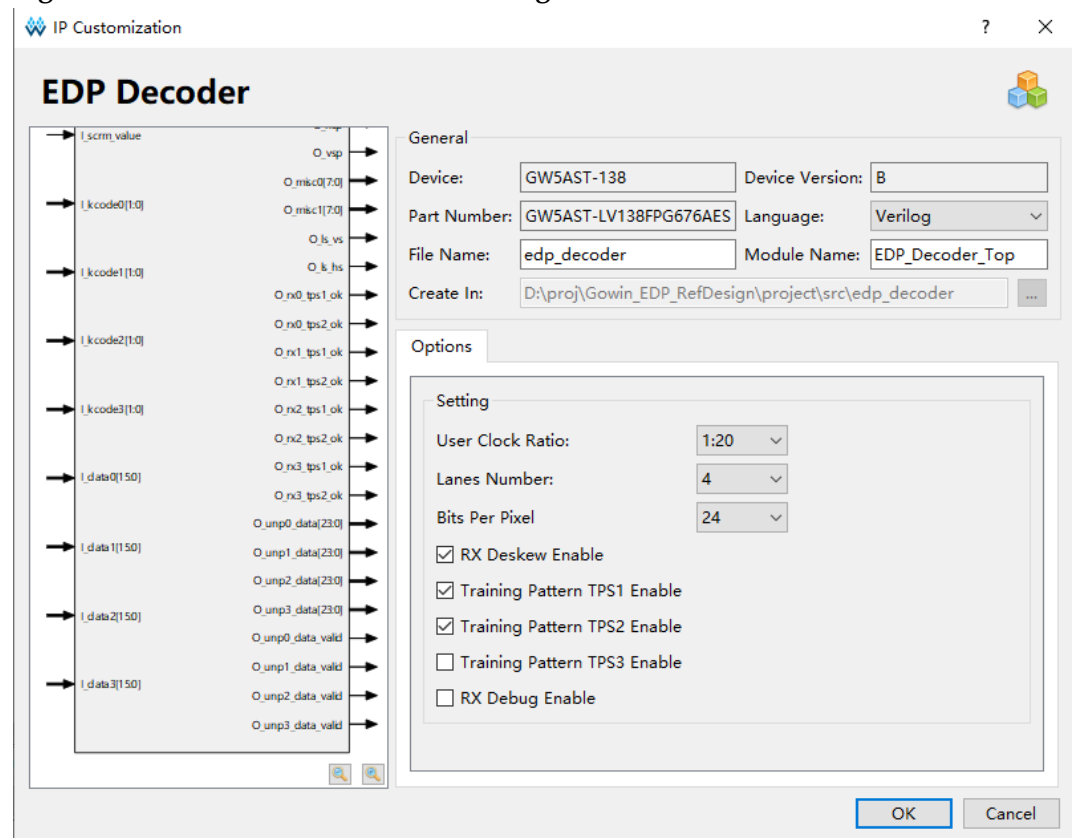
2. Open EDP Decoder IP core

Click "Multimedia" and double click "EDP Decode" to open the configuration interface, as shown in Figure 4-7.

Figure 4-7 Open EDP Decoder IP Core

3. EDP Decoder IP Core Port Interface

On the left of the configuration interface is the port diagram of EDP Decoder IP core, as shown in Figure 4-8.

Figure 4-8 EDP Decoder IP Core Port Diagram

4. Configure General Information

See the project general information in the configuration interface. Take GW5AST-138 and FPG676A package as an example. The "Module Name" displays the top-level file name of the generated project, and the default is "EDP_Decode_Top"; you can modify the name. The "File Name" displays the folder generated by the IP core, which contains the files required by EDP Decoder IP core, and the default is "edp_decode", and you can modify the path. "Create In" displays the path of IP core folder, and the default is "\\project path\\src\\edp_decode"; you can modify the path.

Figure 4-9 General Configuration

The figure shows a 'General' configuration window with the following fields and values:

Device:	GW5AST-138	Device Version:	B
Part Number:	GW5AST-LV138FPG676AES	Language:	Verilog
File Name:	edp_decoder	Module Name:	EDP_Decoder_Top
Create In:	D:\proj\Gowin_EDP_RefDesign\project\src\edp_decoder		

5. Options Tab

In the tab, you can configure EDP Decoder parameters as needed.

Figure 4-10 Options Configuration

The figure shows an 'Options' configuration window with the following settings:

Setting	
User Clock Ratio:	1:20
Lanes Number:	1
Bits Per Pixel	24
☒ RX Deskew Enable	
☒ Training Pattern TPS1 Enable	
☒ Training Pattern TPS2 Enable	
☒ Training Pattern TPS3 Enable	

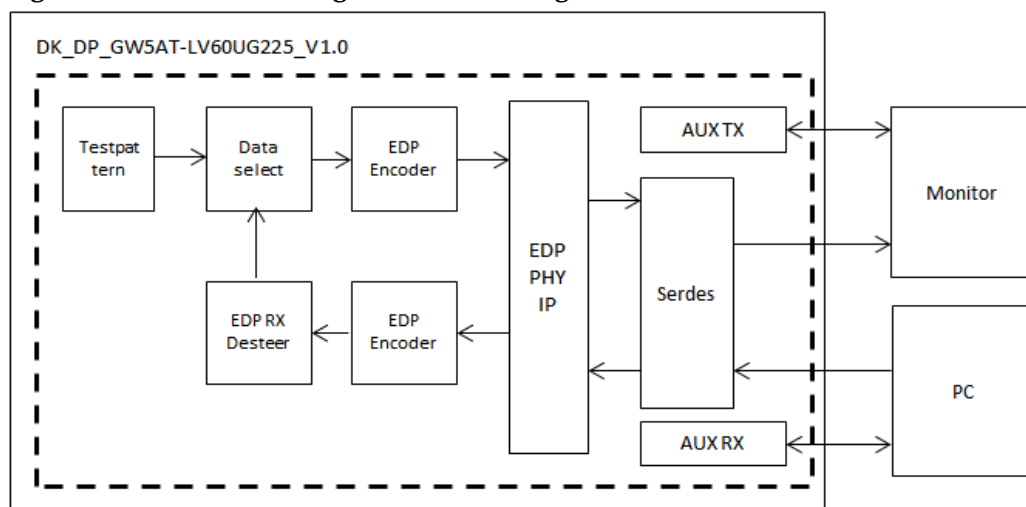
5 Reference Design

This chapter is intended to introduce the usage and structure of the reference design of EDP Encoder Decoder IP. Please see the [reference design](#) for details at Gowinsemi website.

Design Instance

Take DK_CoreBoard_GW5AT-LV138FPG676_V1.0 and DK_DCard_DP-eDP-HDMI-MIPI-LVDS_V1.0 as an example, the diagram is as shown in Figure 5-1. For more information about the development boards, see the details at Gowinsemi website.

Figure 5-1 Reference Design Instance 1 Diagram



In the reference design, there are DP TX and DP RX. The DP TX generates test pattern video signals through Testpattern module; then encode them through EDP Encoder module, input them to EDP PHY IP to convert them into serial signals, and output them to DP display. If no data is received, the DP RX continuously displays the test pattern. For the DP RX, PC outputs DP signals; then convert them to parallel signals through EDP PHY IP, decode them through the EDP Decoder module, convert the format through the EDP RX Desteer module, and then output them to the data selection module before displaying them on the monitor.

6 File Delivery

The file delivery for Gowin EDP Encoder Decoder IP includes the documentation, the design source code, and the reference design.

6.1 Documentation

The folder mainly contains the user guide in PDF version.

Table 6-1 Documents List

Name	Description
IPUG1179, Gowin EDP Encoder Decoder IP User Guide	Gowin EDP Encoder Decoder IP User Guide, i.e., this manual.

6.2 Design Source Code (Encryption)

The encrypted code folder contains encrypted RTL code for Gowin EDP Encoder Decoder IP. This code is intended for use with GUI to generate the IP core as needed.

Table 6-2 Design Source Code List of EDP Encode

Name	Description
edp_encode.v	The top-level file of the IP core, which provides users with interface information, encrypted.

Table 6-3 Design Source Code List of EDP Decode

Name	Description
edp_decode.v	The top-level file of the IP core, which provides users with interface information, encrypted.

6.3 Reference Design

Gowin EDP RefDesign folder contains the netlist file for Gowin EDP Encoder Decoder IP, user reference design, constraints file, top-level file, and project file, etc.

Table 6-4 Gowin EDP RefDesign Folder Content List

Name	Description
test_top.v	The top module of reference design
fpga_project.cst	Project physical constraints file

Name	Description
fpga_project.sdc	Project timing constraints file
key_debounce.v	Debounce module
pll_init.v	PLL initialization module file
rst_gen.v	Reset generation module file
serdes	EDP PHY IP folder
testpattern.v	Test pattern generation module
auxlink	AUX channel communication folder
gowin_pll	PLL IP folder
edp_decoder	EDP receive module folder
edp_encoder	EDP transmit module folder
edp_rx_desteer	EDP RX desteer module folder

